$M = 100$ puts, $P^{BS}(t,s)$ the put price, $\delta_t = \partial_s P^{BS}(t,S_t) =$

$= N(d_1) - 1 < 0$. Per the sheet, scaled by M: $Y_t = 0$, $h_t = M\delta_t$,

$Y_t^{new} = Y_t - h_t S_t = -h_t S_t$

$$V_t = Y_t^{new} + h_t S_t - M P^{BS}(t, S_t) = -M P^{BS}(t, S_t)$$

with $S_{t+\Delta} = S_t e^x$, $x = X_{t+\Delta} \sim N((\mu - \frac{1}{2}\sigma^2)\Delta, \sigma^2\Delta)$,

$$V_{t+\Delta} = h_t S_t e^x + Y_t^{new} e^{r\Delta} - M P^{BS}(t+\Delta, S_t e^x) =$$

$$= h_t S_t (e^x - e^{r\Delta}) - M P^{BS}(t+\Delta, S_t e^x)$$

$$L_{t+\Delta} = V_t - V_{t+\Delta} = M[P^{BS}(t+\Delta, S_t e^x) - P^{BS}(t, S_t)] -$$

$$- M\delta_t S_t (e^x - e^{r\Delta}).$$

where $P^{BS} = 27.11$, $\delta_t = -0.6086$, $h_t = -60.86$ shares

$V_t = -2,710.77$

Q 1

For a constant dollar position $\vartheta \in \mathbb{R}^d$ with linearized losses $L_{t+j\Delta} = -\vartheta^T X_{t+j\Delta}$ and $X_{t+j\Delta}$ i.i.d. $N(0, \Sigma)$

$$L^1_{t+J\Delta} = \sum_{j=1}^{J} L_{t+j\Delta} = -\vartheta^T \sum_{j=1}^{J} X_{t+j\Delta}, \quad \sum_{j=1}^{J} \sim N(0, J\Sigma),$$

$$L^t_{t+J\Delta} \sim N(0, J\vartheta^T\Sigma\vartheta)$$

so $\mathrm{VaR}_\alpha\left(L^{+}_{t+J\Delta}\right) = \sqrt{J\delta^\top \Sigma \delta}\, q_{p^{-1}}(\alpha) = \sqrt{J}\,\mathrm{VaR}_\alpha(L_{t+\Delta})$

The rule needs linear losses, zero mean, independence across days and normality

Q2.

Grid $T_j = t + j\Delta$, $j = 0, \dots, J$ Period 1 is the setup above with $T_0 = t$ At the start of period $j \geq 2$ the hedge is reset to $h_{T_{j-1}} = M\delta(T_{j-1}, S_{T_{j-1}})$ and the share trade is paid from cash:

$$Y^{new}_{T_{j-1}} = Y_{T_{j-1}} + (h_{T_{j-2}} - h_{T_{j-1}})\, S_{T_{j-1}}, \quad Y_{T_j} = Y^{new}_{T_{j-1}} e^{r\Delta}$$

$$V_{T_j} = Y_{T_j} + h_{T_{j-1}} S_{T_j} - M P^{BS}(T_j, S_{T_j}), \quad L_{T_j} = V_{T_{j-1}} - V_{T_j}$$

Rebalancing is self-financing: the value just after the trade $Y^{new}_{T_{j-1}} + h_{T_{j-1}} S_{T_{j-1}} - M P^{BS}$ equals the value just before, $Y_{T_{j-1}} + h_{T_{j-2}} S_{T_{j-1}} - M P^{BS}$, so $V_{T_{j-1}}$ is well defined. The daily losses therefore telescope

$$L^{t}_{t+J\Delta} = \sum_{j=1}^{J} (V_{T_{j-1}} - V_{T_j}) = V_{T_0} - V_{T_J}$$

Q 3

Expand the one day loss to second order about (t, S_t)

$$e^x - 1 = x + \frac{1}{2}x^3 + O(x^3), \quad e^x - e^{r\Delta} = x + \frac{1}{2}x^2 - r\Delta + \dots$$

$$L_{t+\Delta} = M\left[\theta\Delta + \delta S_t\left(x + \tfrac{1}{2}x^2\right) + \tfrac{1}{2}\gamma S_t^2 x^2\right] - M\delta S_t\left[x + \tfrac{1}{2}x^2 - r\Delta\right] + \dots =$$

$$= M\left[\left(\theta + rS_t\delta\right)\Delta + \tfrac{1}{2}\gamma S_t^2 x^2\right]$$

The term linear in x cancels. By the Black Scholes PDE

$$\theta + rS\delta + \tfrac{1}{2}\sigma^2 S^2\gamma = rP^{BS}$$

$$L_{t+\Delta} \approx M_r P^{BS}\Delta + \frac{1}{2}M\gamma S_t^2\left(x^2 - \sigma^2\Delta\right) \approx b\sigma^2\Delta\left(Z^2 - 1\right)$$

$$b = \tfrac{1}{2}M\gamma S_t^2, \quad Z \sim N(0,1)$$

Since $M_r P^{BS}\Delta = 0.01$ is negligible and $x \approx \sigma\sqrt{\Delta}\,Z$.

So assumption (i) fails: the hedged one day loss is not linear but quadratic in the risk factor. centered scaled χ_1^2 with mean ≈ 0 and skewness 2.83.

Daily rehedging resets the delta, so the ten-day loss is a sum of 10 independent copies, $L^t_{t+10\Delta} \approx b\sigma^2\Delta\,(Q - 10)$ with $Q \sim \chi^2_{10}$. Quantile scale with the multiplier so

$$\frac{VaR_{0.95}\left(L^{+}_{t+10\Delta}\right)}{\sqrt{10}\,VaR_{0.95}\left(L_{t+\Delta}\right)} \approx \frac{q_{0.95}\left(\chi^2_{10}\right)-10}{\sqrt{10}\left(q_{0.95}\left(\chi^2_{10}\right)-1\right)} = \frac{18.307-10}{3.162\times 2.841} = 0.925$$

against 0.94 from the simulation.

Assumption (iv) fails and rule overstates 10-day VaR by 6-8%.

The $\sqrt{10}$ rule scales the loss of freshly hedged book, so it implicitly assumes daily re-hedging and lands near that estimate, not frozen one.